

Global CO₂ Emissions

Who Emits, How Much, and What's Changing

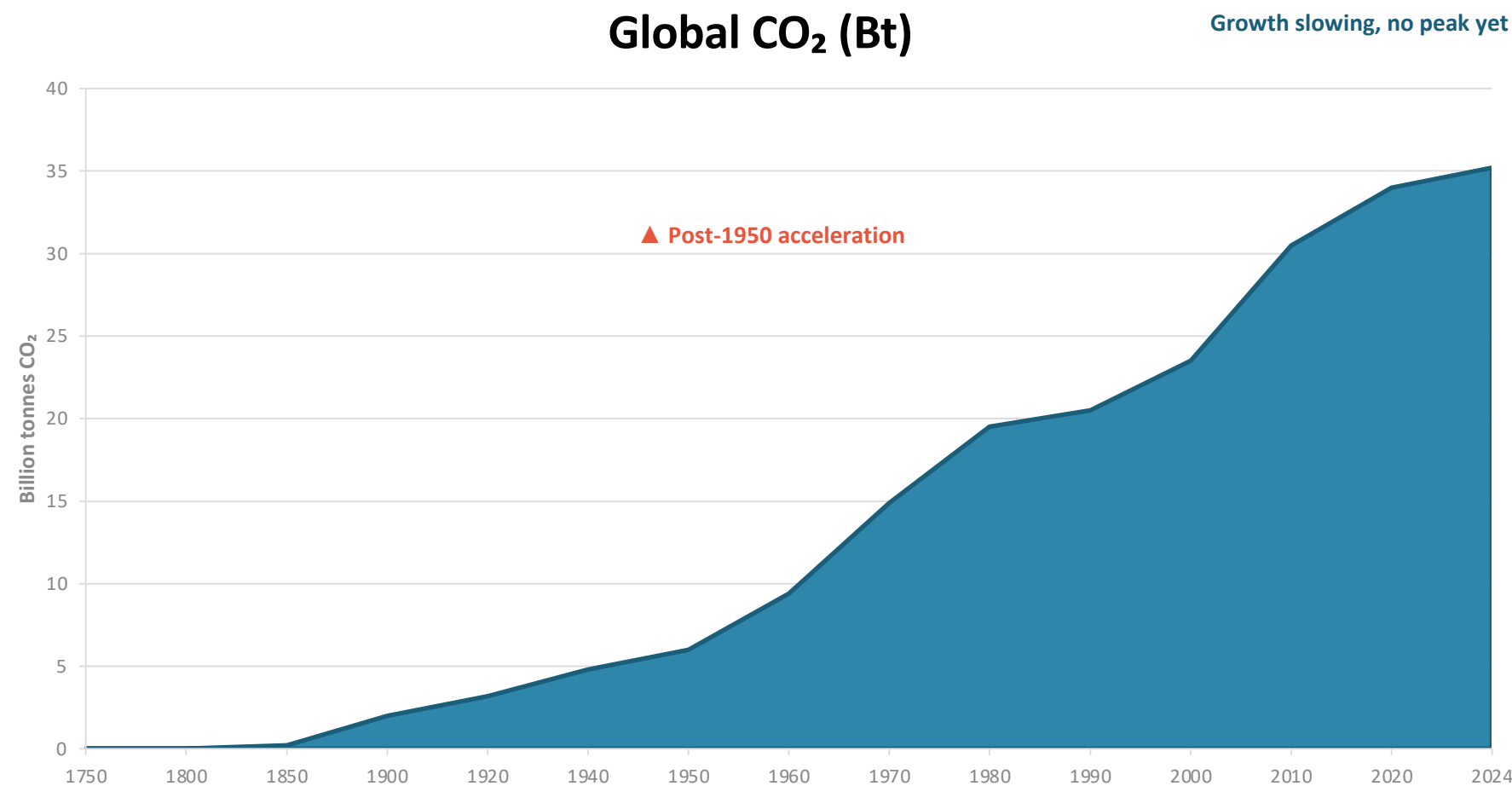
Fossil Fuels & Industry | 1750–2024

Data: Global Carbon Budget 2025 via Our World in Data

For policy analysts, climate negotiators, and sustainability researchers

Global CO₂ Emissions Grew from 6 Bt in 1950 to Over 35 Bt Today

CO₂ emissions from fossil fuels and industry, 1750–2024 (billion tonnes per year)



Key Figures

1750

~0.003 Bt/yr

1950

6 Bt/yr

1990

>20 Bt/yr

2024

>35 Bt/yr

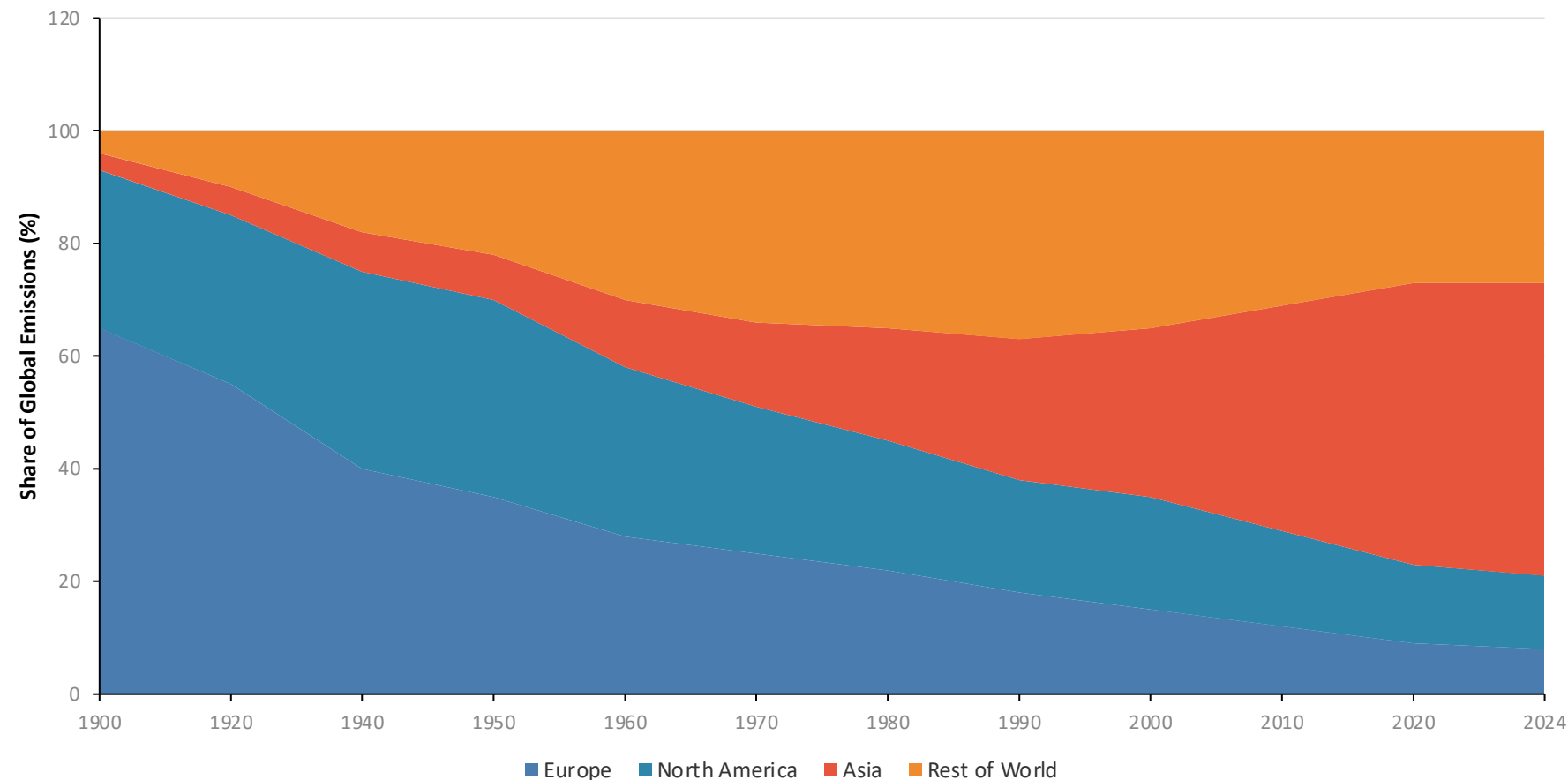
Growth

6× since 1950

Emissions have not yet peaked despite slowing growth

Europe and the US Fell from 90% of Emissions in 1900 to Under One-Third Today

Share of global CO₂ emissions by region, 1900–2024 (stacked area chart)



The Shift

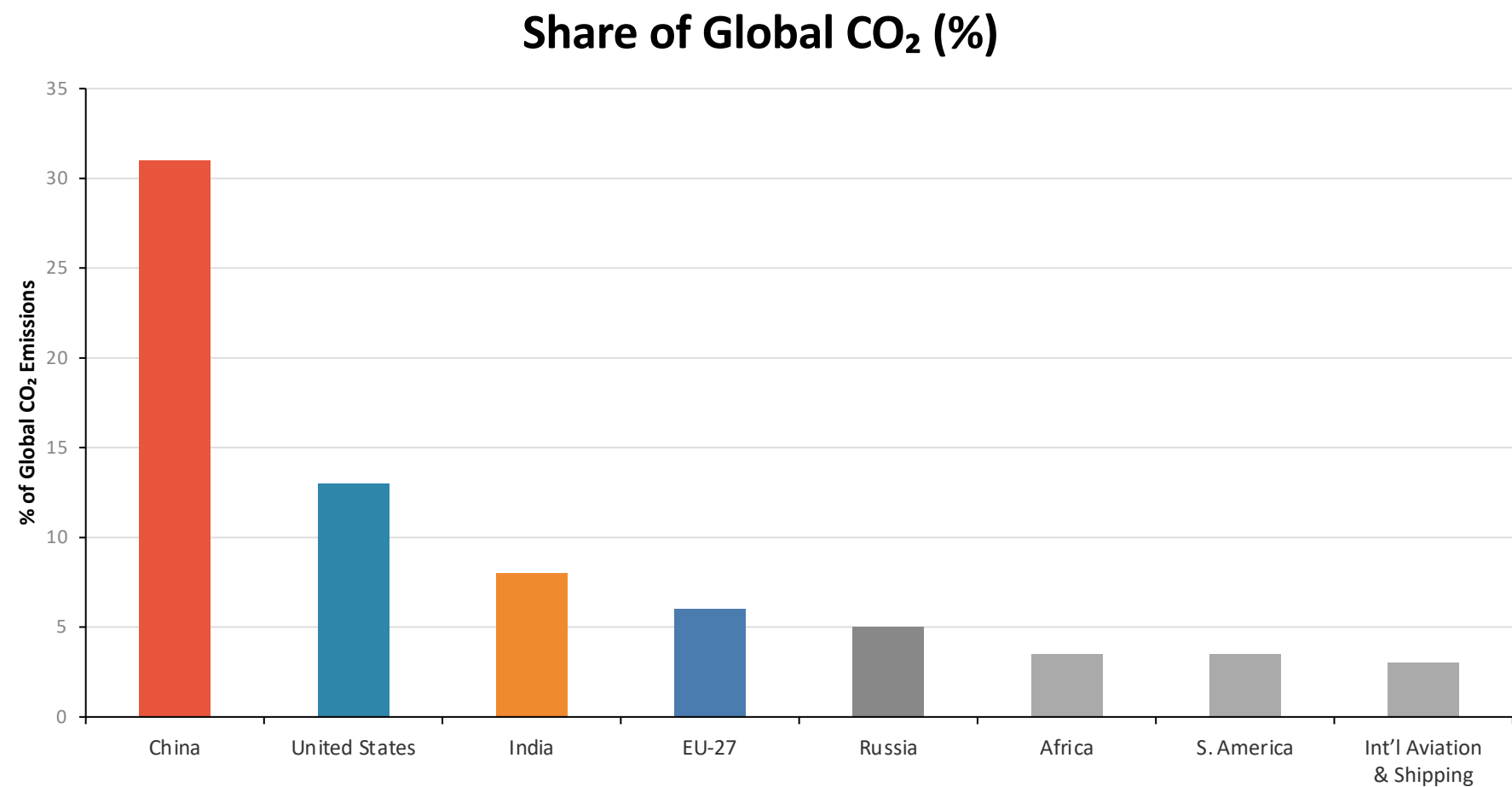
1900: Europe + US = >90% of global emissions

Today: Europe + US = <30%

Asia now produces roughly half of global CO₂, led by China

China Now Emits More Than One-Quarter of Global CO₂

Share of global CO₂ emissions by country/region, 2024



Trends

China ~31%

↑ Rising, growth slowing

US ~13%

↓ Declining

India ~8%

↑ Rising

EU-27 ~6%

↓ Declining

Russia ~5%

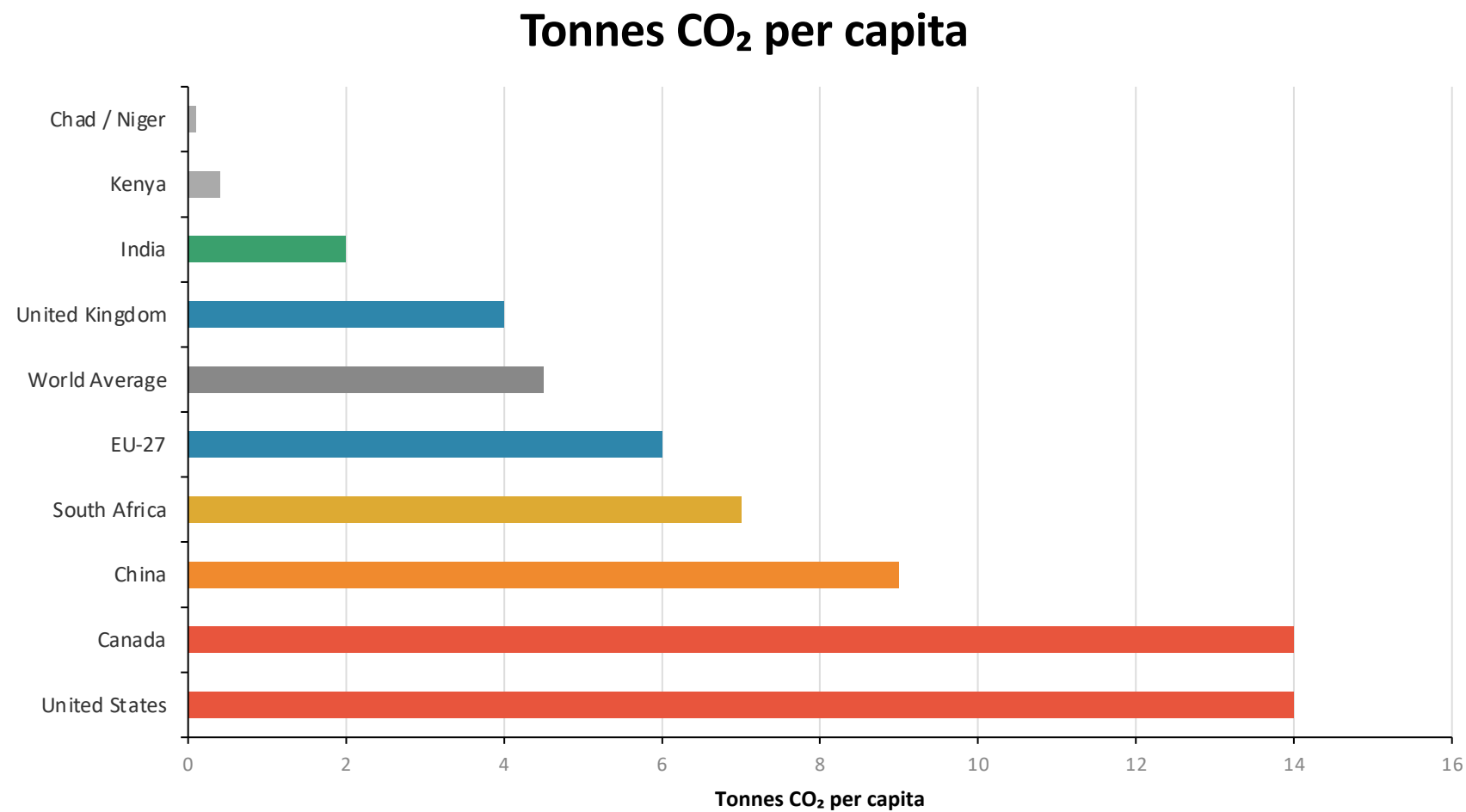
→ Stable

Africa ~3–4%

↑ Slowly rising

Per Capita Emissions Range from 0.1 t in Chad to ~14 t in the US and Canada

Per capita CO₂ emissions by country, 2024 (tonnes per person per year) — a 150× gap



150× GAP

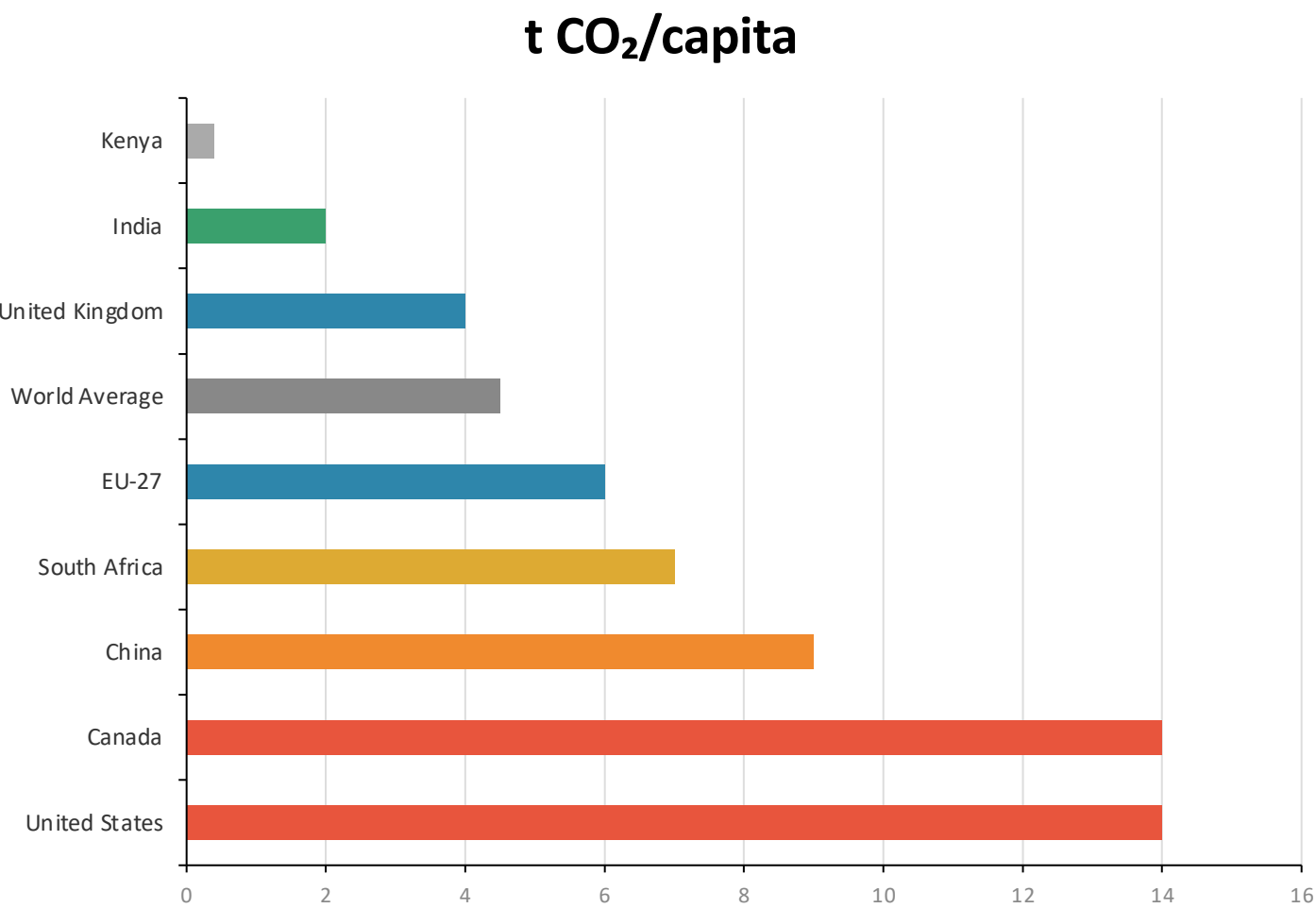
An American or Australian emits in under 2 days what a person in Mali or Niger emits in an entire year.

Prosperity ≠ Emissions

France and the UK have far lower per capita emissions than the US despite similar living standards.

Per Capita Emissions Vary Widely: ~14 t in the US vs. ~9 t in China and ~2 t in India

Per capita CO₂ emissions for key countries, 2024 snapshot (tonnes per person per year)



Key Comparisons

China vs. US

China's per capita (9 t) remains well below the US (14 t) despite being the largest total emitter.

India remains low

At ~2 t per capita, India's emissions are less than half the world average of 4.5 t.

UK vs. US

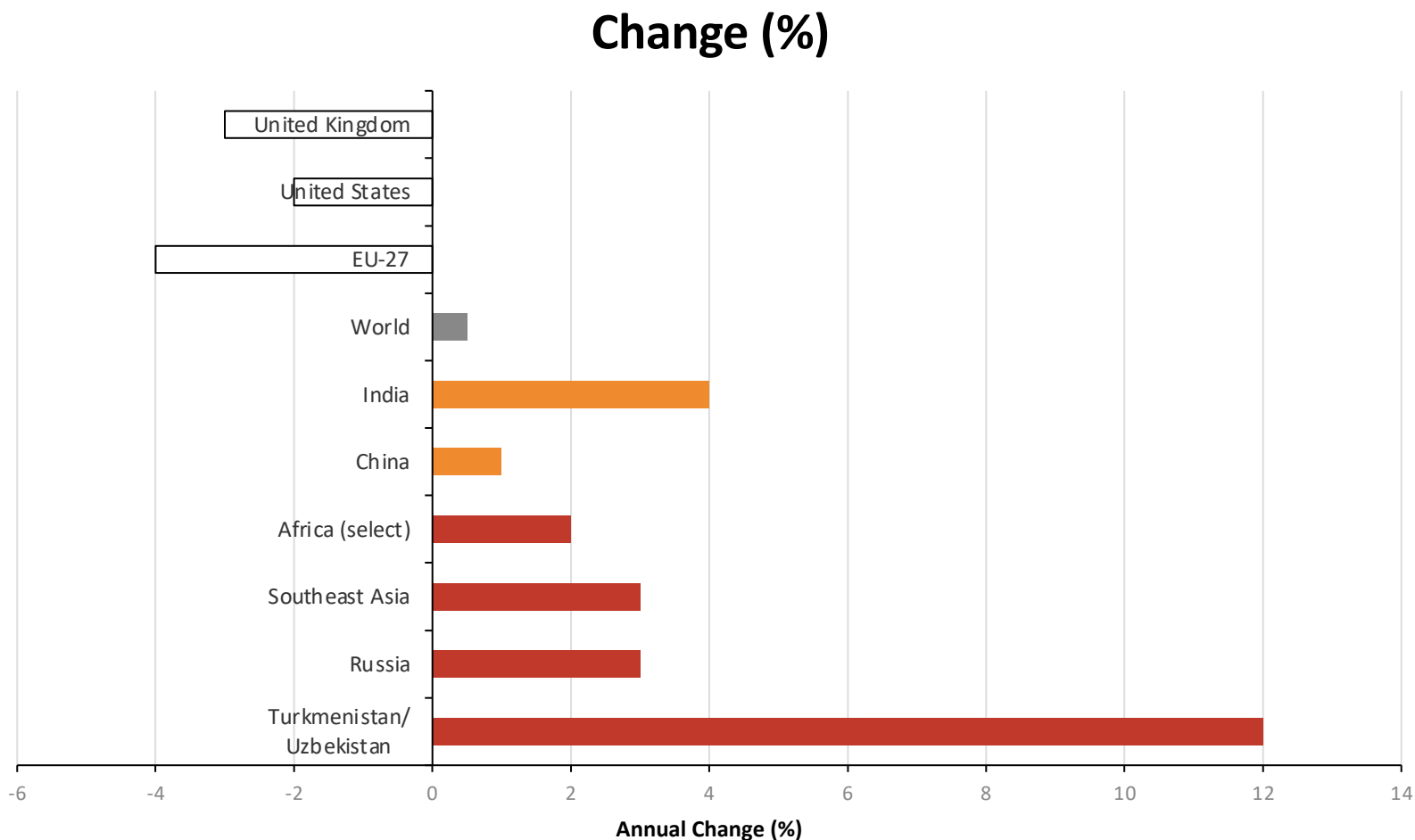
The UK (4 t) emits roughly one-third per capita compared to the US (14 t) with comparable prosperity.

World average

At 4.5 t, the global average is pulled up by a few high-emitting nations; most of the world emits far less.

2024 Snapshot: Many Western Nations Cut Emissions While Parts of Asia and Africa Grew

Annual percentage change in CO₂ emissions, 2024 (diverging bar chart representation)



2024 Signals

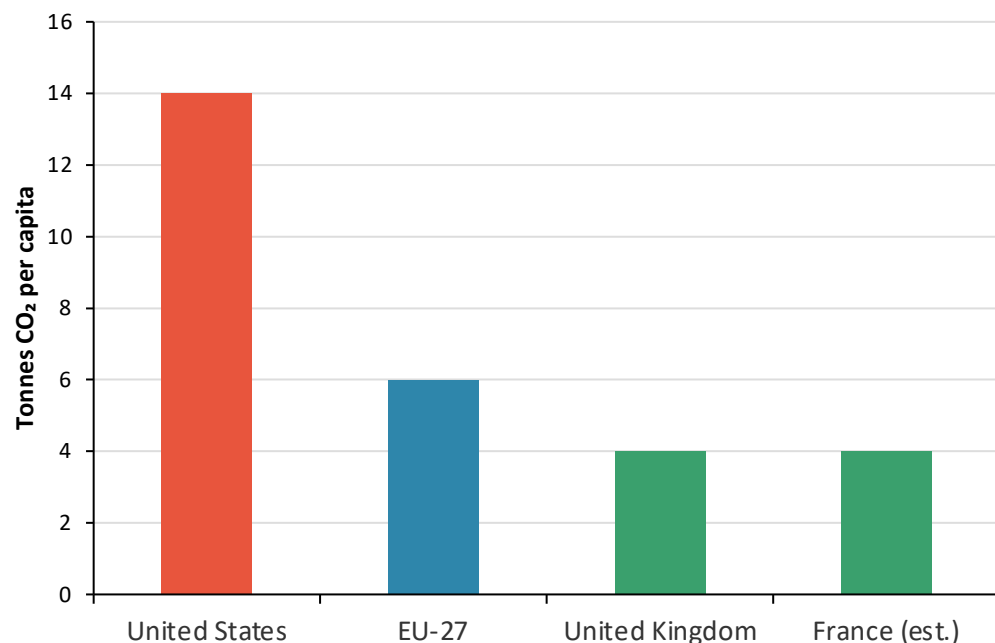
- Declining: US, EU, UK — structural shifts toward clean energy and efficiency
- Increasing: Russia, parts of SE Asia, some African nations — economic growth outpacing clean energy adoption
- Sharp increases: Central Asia (Turkmenistan/Uzbekistan) exceeded 10% growth

Increases in parts of Asia, the Middle East, and Africa offset some of the gains from Western nations' declines.

Energy Mix Explains Why France and the UK Emit Far Less Per Capita Than Germany or the US

Among wealthy nations with similar prosperity, per capita emissions vary significantly due to energy mix choices

Per Capita CO₂ (t)



Why the Difference?

Nuclear & Renewables Lower Emissions

France and the UK have much lower per capita emissions than Germany, the Netherlands, or the US, largely because a higher share of their electricity comes from nuclear and renewable sources rather than fossil fuels.

Same Prosperity, Different Emissions

Countries with similar GDP per capita can have vastly different carbon footprints. The US (~14 t) and UK (~4 t) have comparable living standards but a 3.5× per capita emission gap.

Policy Choices Matter

Energy policy and technology investments can decouple prosperity from emissions — evidence that structural choices, not just wealth, determine carbon intensity.

→ Key Insight: Energy policy and technology choices can decouple prosperity from emissions — France and the UK prove high living standards do not require high per capita emissions.

Key Takeaways: Responsibility Is Shared but Unequal

1

No Peak Yet

Global CO₂ emissions from fossil fuels exceed 35 billion tonnes per year and have not yet peaked, despite slowing growth.

2

China Leads, But Per Capita Tells a Different Story

China is the largest annual emitter (~31% of global CO₂), but its per capita emissions (~9 t) remain below the US (~14 t).

3

Historical Responsibility Falls on Europe and the US

Europe and the US dominated emissions historically (>90% in 1900) but now account for less than one-third of the global total.

4

Massive Per Capita Gaps Persist

A 150× gap exists between the lowest emitters (Sub-Saharan Africa, ~0.1 t/person) and the highest large-country emitters (US/Canada/Australia, ~14–15 t/person).

5

Energy Policy Can Decouple Prosperity from Emissions

Countries with similar prosperity levels can have very different per capita emissions depending on their energy mix — France and the UK emit far less per capita than Germany or the US.